Reliable Machine Learning with Unreliable Black Boxes **Spring 2026**

Lecture 15, part 2: Distribution Shift and Robustness

Thursday, May 21, 2026

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These notes discuss the paper *Discrepancy-Based Active Learning for Domain Adaptation* [de Mathelin, Deheeger, Mougeot, and Vayatis, 2022], as presented by Bob Guo.

1 Setup and Motivation

1.1 Task 1: Domain Adaptation

Train on a source distribution and deploy on a target distribution.

Definition 1 (Domain Adaptation (Covariate Shift))**.** Let Q *(source)* and P *(target)* be distributions over (x, y), with differing x-marginals:

$$Q_X \neq P_X,$$

with labels generated using the same labeling function f, i.e. $y = f(x)$, for both domains.

Goal: Using labeled source samples, learn a function

$$h \in \mathcal{H}$$

with small target risk:

$$\mathcal{L}_P(h, f) = \mathbb{E}_{x \sim P_X}\left[\ell(h(x), f(x))\right]$$

The generalization error can be bounded using a *discrepancy distance* between P and Q.

1.2 Hard Instances for Domain Adaptation

In standard domain adaptation, reweighting source samples can help:

- **Importance sampling:** reweight source samples by

$$\frac{P_X(x)}{Q_X(x)}$$

- **Raking *(Sinkhorn scaling)*:** reweight Q to match (marginal) statistics of P

Problem: What if the target distribution contains regions never seen in the source distribution?
Idea: It would be helpful to obtain a few labeled samples from these unseen target regions.

1.3 Task 2: Active Learning

Many real-world machine learning applications have abundant features but very limited labels. Examples include:

- **Autonomous driving**: videos are cheap, annotations are expensive
- **Materials and drug discovery**: testing true molecular properties is costly

Definition 2 (Active Learning (informal))**.** Given unlabeled target samples

$$\mathcal{T} = \{x_i'\} \sim P_X$$

and a query budget

$$k \ll n,$$

select k samples to label, and train a predictor using those queried labels.

1.4 The Joint Problem: Active Learning for Better Domain Adaptation

Suppose we want to learn on the target distribution P with the following setup:

- As in domain adaptation, we have labeled source samples

$$(x_i, f(x_i)) \sim Q.$$

- As in active learning, we have unlabeled target samples

$$x_j' \sim P_X.$$

  We may query k of these labels.

Definition 3 (Main Question)**.** Which k target points should be queried in order to train a good predictor on P?

Equivalently, this is active learning with a "warm start" set of labeled samples.

1.5 Contributions

1. **Theory:** A domain adaptation generalization bound based on localized discrepancy
2. **Algorithm:** The bound naturally motivates an active learning algorithm
3. **Experiments:** The proposed method outperforms previous baselines

2 Notation and (Localized) Discrepancy

2.1 Formal Setting

The analysis considers empirical distributions:

$$\widehat{Q} \quad \text{over} \quad \mathcal{S} = \{x_i\},$$

and

$$\widehat{P}_X \quad \text{over} \quad \mathcal{T} = \{x'_i\}.$$

Assumptions:

- Deterministic labeling function f
- Hypothesis class $\mathcal{H}$ consisting of $\mathcal{L}_{\mathcal{H}}$-Lipschitz functions
- Loss function

  $$\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$$

  which is:

  - symmetric
  - bounded by M
  - satisfies the triangle inequality
  - $\mathcal{L}_\ell$-Lipschitz

The risk is defined as:

$$\mathcal{L}_D(h, h') = \mathbb{E}_{x \sim D_X}\left[\ell(h(x), h'(x))\right]$$

2.2 A Brief History of Discrepancy Measures

- [Ben-David, Blitzer, Crammer, Kulesza, Pereira, and Vaughan, 2010]: $d_{\mathcal{H}\Delta\mathcal{H}}$ divergence for classification
- [Mansour, Mohri, and Rostamizadeh, 2009]: discrepancy distance

  $$\mathrm{disc}_{\mathcal{H}}(\widehat{P}, \widehat{Q}) = \max_{h,h' \in \mathcal{H}} \left| \mathcal{L}_{\widehat{P}}(h, h') - \mathcal{L}_{\widehat{Q}}(h, h') \right| \quad \text{for any loss } \ell$$

- [Cortes, Mohri, and Medina, 2019]: generalized discrepancy
- [Zhang, Long, Wang, and Jordan, 2020]: localized discrepancy: maximize over all good-fit hypotheses, instead of all of $\mathcal{H}$
- [de Mathelin, Deheeger, Mougeot, and Vayatis, 2022]: localized discrepancy combined with active learning for domain adaptation

2.3 Discrepancy [Mansour, Mohri, and Rostamizadeh, 2009]

The discrepancy asks whether there exists a pair of hypotheses

$$h, h' \in \mathcal{H}$$

that behave similarly on the source distribution but differently on the target distribution.

Definition 4 (Discrepancy Distance *(population)*)**.**

$$\mathrm{disc}_{\mathcal{H}}(\widehat{P}, \widehat{Q}) := \max_{h,h' \in \mathcal{H}} \Big(\mathcal{L}_{\widehat{P}}(h, h') - \mathcal{L}_{\widehat{Q}}(h, h') \Big)$$

For any $h \in \mathcal{H}$, compare its risk against the optimal hypothesis $h^* \in \mathcal{H}$:

$$\begin{aligned} \mathcal{L}_{\widehat{P}}(h, f) &\leq \mathcal{L}_{\widehat{P}}(h, h^*) + \mathcal{L}_{\widehat{P}}(h^*, f) \\ &\leq \mathcal{L}_{\widehat{Q}}(h, h^*) + \mathrm{disc}_{\mathcal{H}}(\widehat{P}, \widehat{Q}) + \mathcal{L}_{\widehat{P}}(h^*, f) \\ &\leq \mathcal{L}_{\widehat{Q}}(h, f) + \mathrm{disc}_{\mathcal{H}}(\widehat{P}, \widehat{Q}) + \mathcal{L}_{\widehat{Q}}(h^*, f) + \mathcal{L}_{\widehat{P}}(h^*, f) \end{aligned}$$

Define

$$\eta_\infty := \min_{h^* \in \mathcal{H}} \max_{x \in \mathcal{S} \cup T} \{\ell(h^*(x), f(x))\}$$

Then:

$$\mathcal{L}_{\widehat{P}}(h, f) \leq \underbrace{\mathcal{L}_{\widehat{Q}}(h, f)}_{\text{source risk}} + \underbrace{\mathrm{disc}_{\mathcal{H}}(\widehat{P}, \widehat{Q})}_{\text{discrepancy}} + \underbrace{\eta_\infty}_{\substack{\text{approx. error} \\ (= 0 \text{ if } f \in \mathcal{H})}}$$

Note that $f \in \mathcal{H} \implies \eta_\infty = 0$

2.4 From Regular to Localized Discrepancy

Standard discrepancy is pessimistic because it considers all pairs of hypotheses in $\mathcal{H}$, including irrelevant ones.

Definition 5 (Localized Discrepancy [Zhang, Long, Wang, and Jordan, 2020])**.** Restricting to hypotheses with max error $\leq \epsilon$ on *source* samples:

$$\mathcal{H}_\epsilon := \{h \in \mathcal{H} : \ell(h(x), f(x)) \leq \epsilon \quad \forall x \in \mathcal{S}\}$$

Define:

$$\mathrm{disc}_{\mathcal{H}_\epsilon}(\widehat{P}, \widehat{Q}) = \max_{h,h' \in \mathcal{H}_\epsilon} \Big(\mathcal{L}_{\widehat{P}}(h, h') - \mathcal{L}_{\widehat{Q}}(h, h') \Big)$$

- Smaller ϵ gives tighter bounds
- We require $\epsilon \geq \eta_\infty$ so that $\mathcal{H}_\epsilon$ is nonempty
- Ideally, $\mathcal{H}_\epsilon$ should be small while still containing h^*

3 Main Result and Proof Sketch

3.1 Upper Bounding Discrepancy

We begin with:

$$\mathcal{L}_{\widehat{P}}(h,f) \leq \underbrace{\mathcal{L}_{\widehat{Q}}(h,f)}_{\text{source risk}} + \underbrace{\mathrm{disc}_{\mathcal{H}_\epsilon}(\widehat{P},\widehat{Q})}_{\text{discrepancy}} + \underbrace{\eta_\infty}_{\substack{\text{approx. error} \\ (=0 \text{ if } f\in\mathcal{H})}}$$

Goal of Active Learning: Query points in T and add them to S in order to reduce $\mathrm{disc}_{\mathcal{H}_\epsilon}(\widehat{P},\widehat{Q})$.

Problem: The discrepancy is a maximization over all function pairs, hard to compute directly.
Idea: Use the Lipschitz assumptions on $\mathcal{H}$ and ℓ.

3.2 Using Lipschitzness to Bound Discrepancy

Consider source $x_0 \in \mathcal{S}$ and target $x_1 \in \mathcal{T}$ samples, which satisfy:

- $\forall h \in \mathcal{H}_\epsilon, \quad \ell(h(x_0), f(x_0)) \leq \epsilon$
- h can drift by at most $\mathcal{L}_\mathcal{H}\|x_1 - x_0\|$
- $\ell(h(x_1), f(x_0)) \leq \mathcal{L}_\ell \mathcal{L}_\mathcal{H}\|x_1 - x_0\| + \epsilon$ *(by the Lipschitz property of the loss and the triangle inequality)*

Thus:

$$\begin{aligned}
\ell(h(x_1), h'(x_1)) &\leq \ell(h(x_1), f(x_0)) + \ell(h'(x_1), f(x_0)) \\
&\leq 2\left(\mathcal{L}_\ell \mathcal{L}_\mathcal{H}\|x_1 - x_0\| + \epsilon\right)
\end{aligned}$$

Averaging over $x' \in \mathcal{T}$:

$$\mathrm{disc}_{\mathcal{H}_\epsilon}(\widehat{P},\widehat{Q}) \leq \boxed{\frac{2\mathcal{L}_\mathcal{H}\mathcal{L}_\ell}{|\mathcal{T}|}\sum_{x'\in\mathcal{T}} d(x', S) + 2\epsilon}, \quad d(x', S) = \min_{x\in\mathcal{S}} \|x' - x\|$$

3.3 Main Result

Theorem 6 ((informal)). *Under the assumptions above, for all $\epsilon \geq \eta_\infty, \quad h \in \mathcal{H}_\epsilon$:*

$$\mathcal{L}_{\widehat{P}}(h,f) \leq \mathcal{L}_{\widehat{Q}_k}(h,f) + \boxed{\frac{2\mathcal{L}_\mathcal{H}\mathcal{L}_\ell}{|\mathcal{T}|}\sum_{x'\in\mathcal{T}} d(x', S) + 2\epsilon} + \eta_\infty$$

The paper further derives a bound for the population risk.

Theorem 7. *For any $\epsilon \geq \eta_\infty, h \in \mathcal{H}_\epsilon$, with probability at least $1-\delta$,*

$$\mathcal{L}_P(h,f) \leq \underbrace{\mathcal{L}_{\widehat{Q}_k}(h,f)}_{\textit{fit}} + \underbrace{\frac{2\mathcal{L}_\mathcal{H}\mathcal{L}_\ell}{|\mathcal{T}|}\sum_{x'\in\mathcal{T}} d(x',S)}_{\textit{coverage}}$$

$$+ \underbrace{2\epsilon+\eta_\infty}_{\textit{approximation}} + \underbrace{2\mathcal{L}_\ell\mathcal{R}_{|\mathcal{T}|}(\mathcal{H}) + \sqrt{\frac{M^2\log(1/\delta)}{2|\mathcal{T}|}}}_{\substack{\textit{statistical error} \\ \textit{(estimating } \mathcal{L}_P \textit{ from } \mathcal{L}_{\hat{P}}\textit{)}}}$$

3.4 Interpretation

The error decomposition becomes:

$$\mathcal{L}_{\widehat{P}}(h,f) \leq \underbrace{\mathcal{L}_{\widehat{Q}_k}(h,f)}_{\text{fit}} + \underbrace{\frac{2\mathcal{L}_\mathcal{H}\mathcal{L}_\ell}{|\mathcal{T}|}\sum_{x'\in\mathcal{T}} d(x',S)}_{\text{coverage}} + \underbrace{2\epsilon+\eta_\infty}_{\text{approximation}}.$$

Caveats:

- Requires the true labeling function f to be close to $\mathcal{H}$ and deterministic
- Must choose
$$\epsilon \geq \max_{x\in\mathcal{S}} \ell(f(x),h(x)) \geq \eta_\infty$$
In practice, selecting ϵ is unclear

Good news: Minimizing the coverage term yields a clean active learning objective. This becomes exactly the k-medoids problem, except with initial medoids fixed.

4 Algorithm

4.1 From the Bound to an Algorithm

Definition 8 (Active Learning Objective)**.**

$$\mathcal{T}_k^* = \arg\min_{\mathcal{T}_k\subseteq T,\ |\mathcal{T}_k|=k} \sum_{x'\in\mathcal{T}} d(x', S\cup\mathcal{T}_k)$$

- This is a k-medoids problem on $\mathcal{T}$ with source S as pre-existing fixed medoids
- The optimization problem is NP-hard
- A natural solution is to use a greedy heuristic

4.2 Greedy k-Medoids

For any queried subset $\mathcal{T}_k \subseteq T$, define:

$$F(\mathcal{T}_k) := \sum_{x' \in \mathcal{T}} \left[d(x', S) - d(x', S \cup \mathcal{T}_k) \right]$$

This measures how much the queried set reduces the coverage objective.

Greedy Algorithm

- Initialize: $\mathcal{T}_0 := \emptyset$
- For $i = 1, \ldots, k$, select:

$$x_i^* \leftarrow \arg \max_{x \in \mathcal{T} \setminus \mathcal{T}_{i-1}} \underbrace{F(\mathcal{T}_{i-1} \cup \{x\}) - F(\mathcal{T}_{i-1})}_{\text{marginal gain from adding } x}$$

- Then update:

$$\mathcal{T}_i \leftarrow \mathcal{T}_{i-1} \cup \{x_i^*\}$$

- Repeat.

At every round, the algorithm greedily decreases the objective as much as possible.
Each round contributes $\mathcal{O}(|\mathcal{T}|)$ marginal-gain evaluations, for a total runtime of $\mathcal{O}(k|\mathcal{T}|^2)$

4.3 Why Does Greedy Work?

The function $F(\mathcal{T}')$ satisfies two key properties.

- **Monotonicity:** $\mathcal{T}_1 \subseteq \mathcal{T}_2, \implies F(\mathcal{T}_1) \leq F(\mathcal{T}_2)$
- **Submodularity:** The marginal gain from adding a point decreases as the selected set grows:

$$F(\mathcal{T}_1 \cup \{x\}) - F(\mathcal{T}_1) \geq F(\mathcal{T}_2 \cup \{x\}) - F(\mathcal{T}_2) \qquad \text{when} \quad \mathcal{T}_1 \subseteq \mathcal{T}_2$$

Theorem 9 ([Nemhauser, Wolsey, and Fisher, 1978])**.** *For monotone submodular maximization under the constraint $|\mathcal{T}'| \leq k$:*

$$F(\mathcal{T}_k^{\mathit{greedy}}) \geq \left(1 - \frac{1}{e}\right) F(\mathcal{T}_k^*)$$

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